PHYSICS 101 · SECTION BSCS 2-A

Lab Report 2: Projectile Motion

Instructor: Mr. Santos · 10 points

Due: Sunday, October 11, 2026, 11:59 PM

Objective

Measure the horizontal range of a projectile launched at three different angles and compare the measured range to the range predicted by the projectile motion equations.

Materials

1. Launcher or ramp (provided in lab)
2. Steel ball, 2 cm diameter
3. Meter stick and stopwatch
4. Protractor

Procedure

1. Set the launch angle to 15°. Record the launch speed printed on the launcher.
2. Release the ball and measure the horizontal distance to first impact. Repeat 3 times and average.
3. Repeat steps 1–2 for launch angles 30° and 45°.
4. For each angle, compute the predicted range using $R = (v^2 \sin 2\theta) / g$ and compare to your measured average.

Data table to submit

Angle	Trial 1 (m)	Trial 2 (m)	Trial 3 (m)	Average (m)	Predicted (m)
15°					
30°					
45°					

Discussion questions

1. At which angle was the measured range closest to the predicted range? Suggest one source of error.
2. Which angle gave the longest range, and does this match theory?

Submit a single PDF with your data table and answers through the LMS. Late submissions lose 10% per day.